

# Environmental Analysis

Al Safa 2 Park — Ecology, Water, Vegetation, Soil, Stormwater & Noise

## 1. Ecology & Biodiversity

- Existing tree canopy (visible west portion of site) is the primary existing ecological asset — mature trees in Dubai’s residential parks are typically Ghaf (Prosopis cineraria - UAE national tree), Neem, Ficus, and date palm plantings; species cannot be confirmed without the DWG planting layer or a site visit.
- No water bodies visible on site — no aquatic habitat present currently.
- As an isolated urban green patch, the park likely functions as a stepping-stone habitat for birds and urban-tolerant species rather than a primary habitat corridor.

## 2. Water

No existing water features identified. Dubai’s water context (Climate Analysis, Phase 1.05): near-zero rainfall Jun-Sep, light rainfall Dec-Feb (~15-23mm/month) — any landscape strategy must assume irrigation-dependent planting, reinforcing the brief’s call for water-sensitive design and sustainable irrigation.

## 3. Vegetation

Visible canopy concentration = west side of site; east side reads more open/hardscape in the aerial graphic. This asymmetry is a genuine existing condition worth carrying into Phase 2 as a potential shade-equity issue.

## 4-8. Soil, Stormwater, Carbon, Habitat, Air, Noise, Visual Corridors

| Aspect           | Status                                                                                                                                           |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Soil             | No geotechnical data provided; regional assumption only (calcareous sandy soils typical of Dubai) — logged as gap                                |
| Stormwater       | No infrastructure data available (sits in DWG utilities layer, pending conversion)                                                               |
| Carbon           | No baseline data; would need modeling once species/quantities known                                                                              |
| Habitat          | Likely supports common urban bird species; no formal survey data                                                                                 |
| Noise            | Sheikh Zayed Road (E11) borders east edge — confirmed major noise source requiring acoustic buffering                                            |
| Air              | No air quality monitoring data for this specific site; SZR proximity is a qualitative concern                                                    |
| Visual Corridors | No confirmed skyline sightline data for this specific site; internal circular plaza appears to be a potential focal point (unconfirmed function) |

## 9. Summary of Data Status

Most items above are qualitative/contextual reads from the one available graphic plus regional climate knowledge. Quantitative/scientific data (soil tests, stormwater capacity, noise dB readings, carbon baselines, biodiversity surveys) require sources not included in the competition package and are logged as gaps in 00\_MASTER\_TRACKER.